

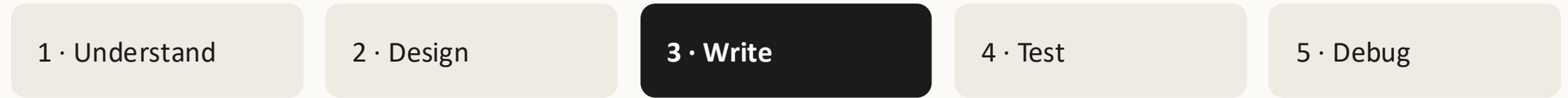
Extension module

Estimating & Planning with AI

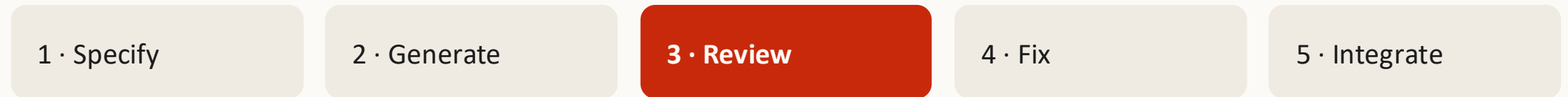
What changes when an agent writes the code?

The cost moved

Without AI



With AI



Before, the effort was in writing. Now it is in specifying, reviewing and integrating.

Three levels of AI involvement

L0 • SE 1.0

Manual

You: understand, design, write, test, debug.

AI: nothing.

Time goes to: writing and debugging.

L1 • SE 2.0

AI-assisted

You: drive every step; ask, read, edit.

AI: suggests code, answers questions.

Time goes to: asking, reading and fixing suggestions.

L2 • SE 3.0

Agentic

You: write the goal and the tests; review.

AI: plans, edits, runs and iterates.

Time goes to: specifying and reviewing.

Vibe coding is **not** a level.

It means accepting code you have not read and cannot explain. Fine for a throwaway prototype. Never for something you ship.

You are accountable for every line your team hands in.

What stays, what changes

Stays the same	Changes
Story points measure size	Points finished per sprint (velocity)
Relative beats absolute estimation	What makes a story risky
Value / Estimate $\times$ 1/Risk ranks the backlog	Which phases you must estimate
Velocity limits the sprint scope	How you learn velocity: measure it

Easy to generate, hard to judge

What you see

The code appears in seconds and it looks right.

What you still have to do

Messy or missing data

Edge cases

Proof that it is right

Integration with the rest

Security and privacy

Running from a clean start

Beginners miss most of it, because they have never done it.

We feel faster than we are

56%

faster at one isolated task

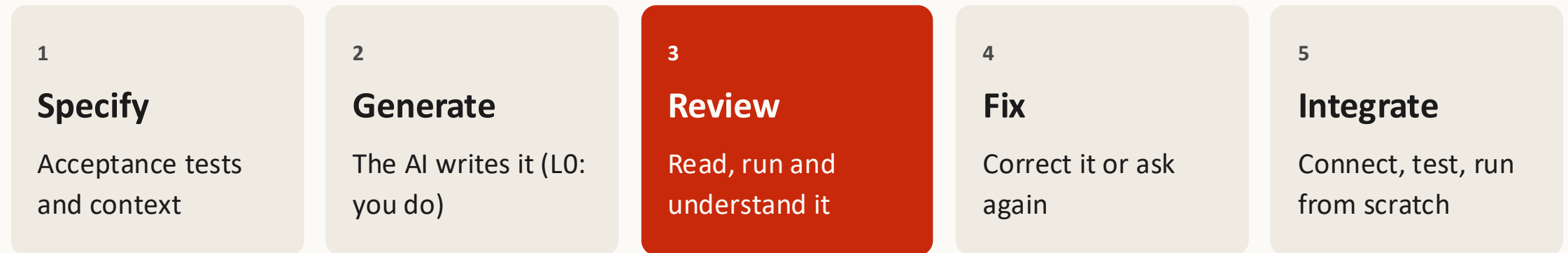
A controlled experiment with GitHub Copilot ([Peng et al., 2023](#)).

19%

slower on real tasks

Experienced developers in their own repositories (METR, 2025). They believed they were about 20% faster.

Estimate the **whole** cycle



Estimate all five phases, not only Generate.

In data science, Review includes checking the data and the metric.

Risk includes **verification**

Easy to verify

A function with clear inputs and outputs. Unit tests say pass or fail.

Example: search by title.

Hard to verify

A model whose metric looks good. Data leakage still runs and prints a number.

Example: personalized recommendations.

Ask: if this were wrong, how would we find out?

Risk (1–5) = the highest of technical uncertainty, ambiguity and verification difficulty.

The user story as a spec

As a <role>
I want <goal>
So that <benefit>

Acceptance criteria (testable, written first):
...

Out of scope: ...

Business Value: XX **Story Points:** X

Risk: X **AI mode:** L0 / L1 / L2

Criteria you can run as tests. If you cannot write them, the story is not ready.

Out of scope. Tells the agent, and the team, what not to touch.

Risk and AI mode. Decided by the team, not by the tool.

Planning poker, upgraded

Before voting

Interrogate

Answer five questions and write the answers down.

Round 1

Size

Vote story points. Silent reveal. No AI.

Round 2

Risk and mode

Score Risk (1–5). Choose L0, L1 or L2.

Vote on size, not on speed.

A story is the same size whether you type it or an agent generates it.

Five questions before you vote

Topic	Ask	Look for
Inputs	What data does it need?	Missing values, duplicates, messy labels
Edge cases	What could go wrong?	Name three unusual or failing cases
Proof	How will we check it?	At least one acceptance test
Connections	What else does it touch?	Other stories, tables, notebooks, APIs
Unknowns	What do we not know yet?	A lot? Vote ? and investigate first

Anchor on a **reference** story

Pick one story everyone understands. Call it a 3.

1 or 2

Smaller than the
reference

3

The reference story

5 or 8

Bigger than the
reference

13 or more

Too big: split it

After Sprint 1, replace it with a story whose real effort you measured.

Measure velocity, don't assume it

When	Rule
Sprint 1	$1/3 \times (\text{members} \times \text{days})$, with no AI bonus. Example: 4 people $\times$ 6 days = 24, so 8 points.
During the sprint	Log hours per phase and per mode (L0, L1, L2).
Sprint 2	Use the velocity you measured, by mode if the data allows it.

There is no fixed AI discount: the effect depends on the story and on the team.

Timebox the agent in L2

Set the limit before you start, for example 3 attempts or 30 minutes.

If it has not converged when the limit is reached, choose one:

Rewrite the spec

The story or its tests were unclear.

Split the story

It was bigger than you thought.

Go down a level

Switch to L1 or L0 and learn it.

Definition of **Done** with AI

1. Someone on the team can explain every line.
2. The acceptance tests pass, and you have read the tests too.
3. No invented libraries, functions or data columns.
4. It runs from a clean start: restart the notebook, same seed.
5. No secrets, personal data or unlicensed code in the repository.
6. The task card says how AI was used.

AI makes code cheap to **produce**. Understanding it, checking it and owning it are still yours to estimate.

Estimate size. Choose the mode. Measure what happens.